The Economic Impact and Market Forces of Shellfish Harvesting

September Project Update

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Gulf of Maine Research Institute

Introduction and Aims

Increase understanding of landings, ex-vessel prices, and additional factors impacting Maine soft-shell clam fisheries and markets

Topic 1: Understanding values and interactions associated with the soft-shell clam

Topic 2: Factors that affect soft-shell clams, shellfish price structures and pricing fluctuations

Goals of Project

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Provide Economic Analysis and Information to Support a Better Understanding of Soft-Shell Clam Landing Price Trends and Fluctuations at the Municipal, County, Regional, and State Levels

Literature Review of Current Studies

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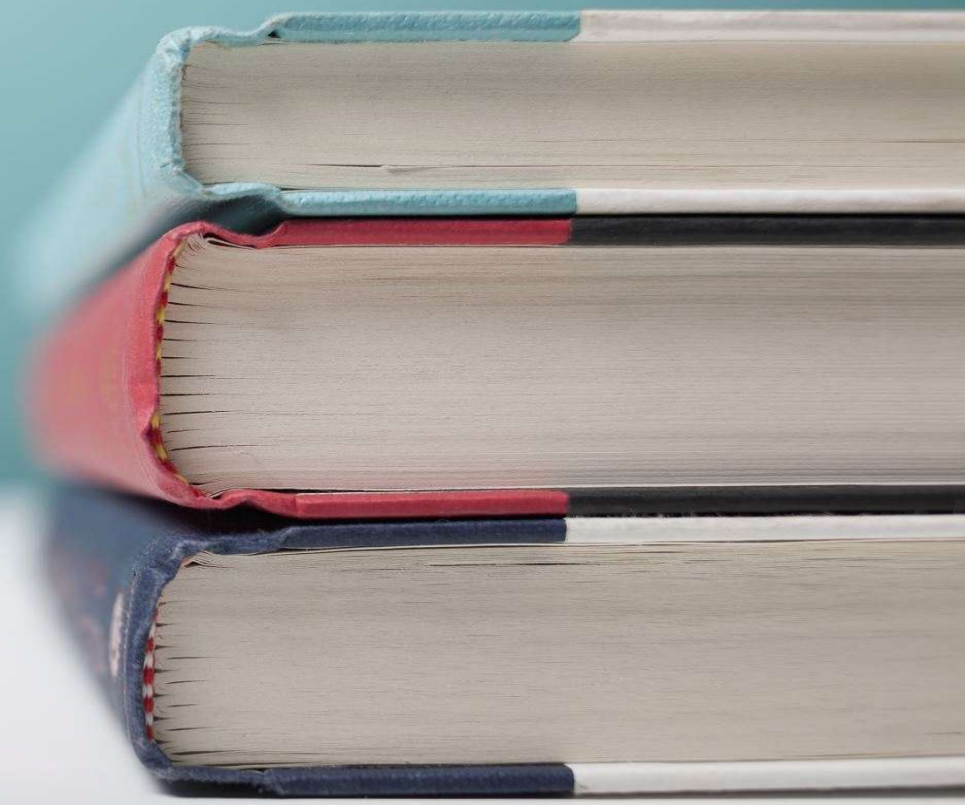
Conducted a Systematic Review of Academic Literature Discussing Soft-Shell Clams and Related Product Markets

Collected further Technical Reports and Additional Documents through website scanning from local and regional resources



- Identified 30+ Journal Articles, Technical Reports, and Other Management-Related Documents

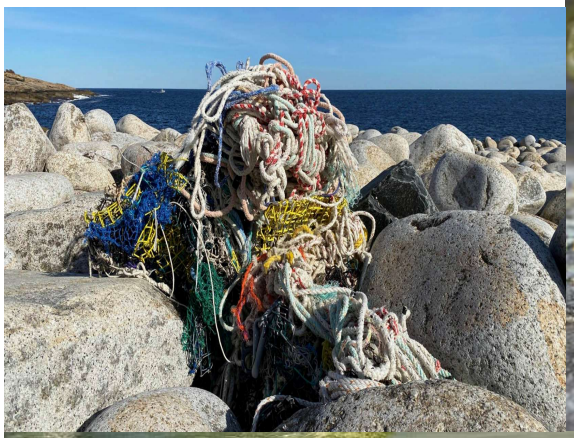
Key Literature Themes

Environmental Factors

- Green Crab Predation, Harmful Algal Blooms, and Pollution are three factors consistent problems for soft-shell clams in Maine

- Sea temperature increases and water quality are additionally affecting soft-shell clams in Maine

- Economic Losses from Pollution Closures can be significant in Maine
  - Temporary Pollution Closures from 2001-2009 led to 27.4% loss in forgone revenue (Evans et al., 2016)



Key Literature Themes

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Socio-Economic Factors

- Coastal Access challenges are becoming more prevalent for Maine clammers
  - Changes in ownership have resulted in retractions of previously held agreements between landowners and clammers (Zoellick et al., 2023)
  - Documented work with 6 Mid-Coast towns found 65% of intertidal access areas were on private property (Farr & McMahon, 2023)

- New Management Options for Municipalities are Increasing in Maine
  - Co-Managed clam fisheries are shown to have a higher degree of stability and more consistent productivity (McClenachan et al., 2015)
  - Increase of shellfish stock enhancement in Maine (Beal, 2002; Tan & Beal, 2015)

Key Literature Themes

Macroeconomic Effects in Maine

- Maine shellfish remains among the state's most valuable fishery
  - Soft-Shell Clams netted Maine harvesters $16 million in 2022 (DMR, 2023)

- Little literature talked about the effects of the COVID-19 pandemic (Smith et al., 2020; Farr & McMahon, 2023)
  - We are interested in understanding this further in this study

- Literature suggests a strong market for Maine shellfish among United States consumers (Athearn, 2008)

Initial Analysis of Data

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Utilizing Maine Department of Marine Resources Commercial Fisheries Landings and Ex-Vessel Values Data

- Obtained data aggregated at the weekly port level from 2006-2023
- Data will be examined from both a state and regional level (Southern, Mid-Coast, and Downeast)
  - **We are interested in ShAC's thoughts on aggregation levels**

Initial Findings

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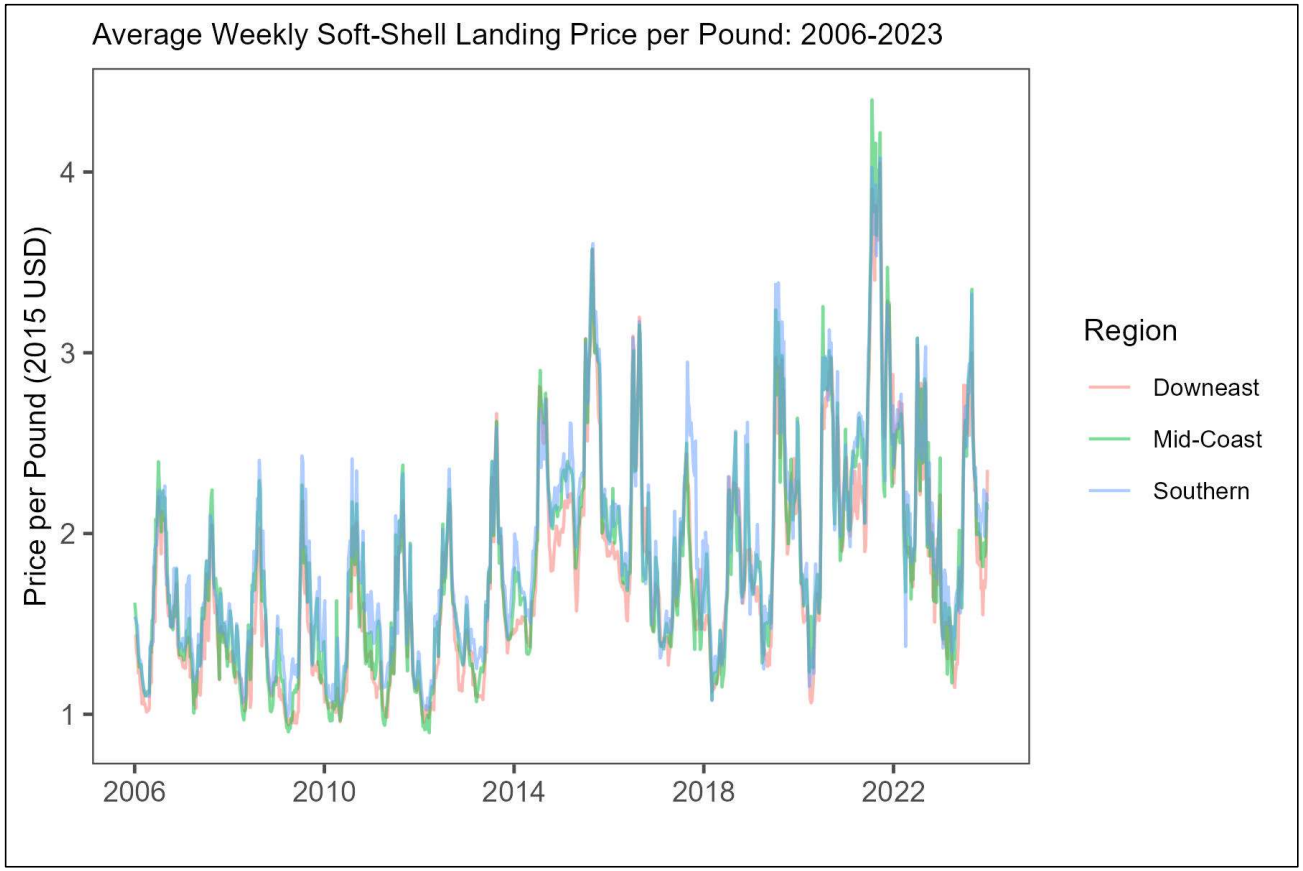
- Prices have been deflated relative to 2015 USD
- 2015 appears to shift prices
  - Mean Price **2006-2015**
    - $1.62 (S.D. of $0.54)
  - Mean Price **2016-2023**
    - $2.09 (S.D. of $0.68)
- Prices appear to become more volatile after 2015



Initial Findings

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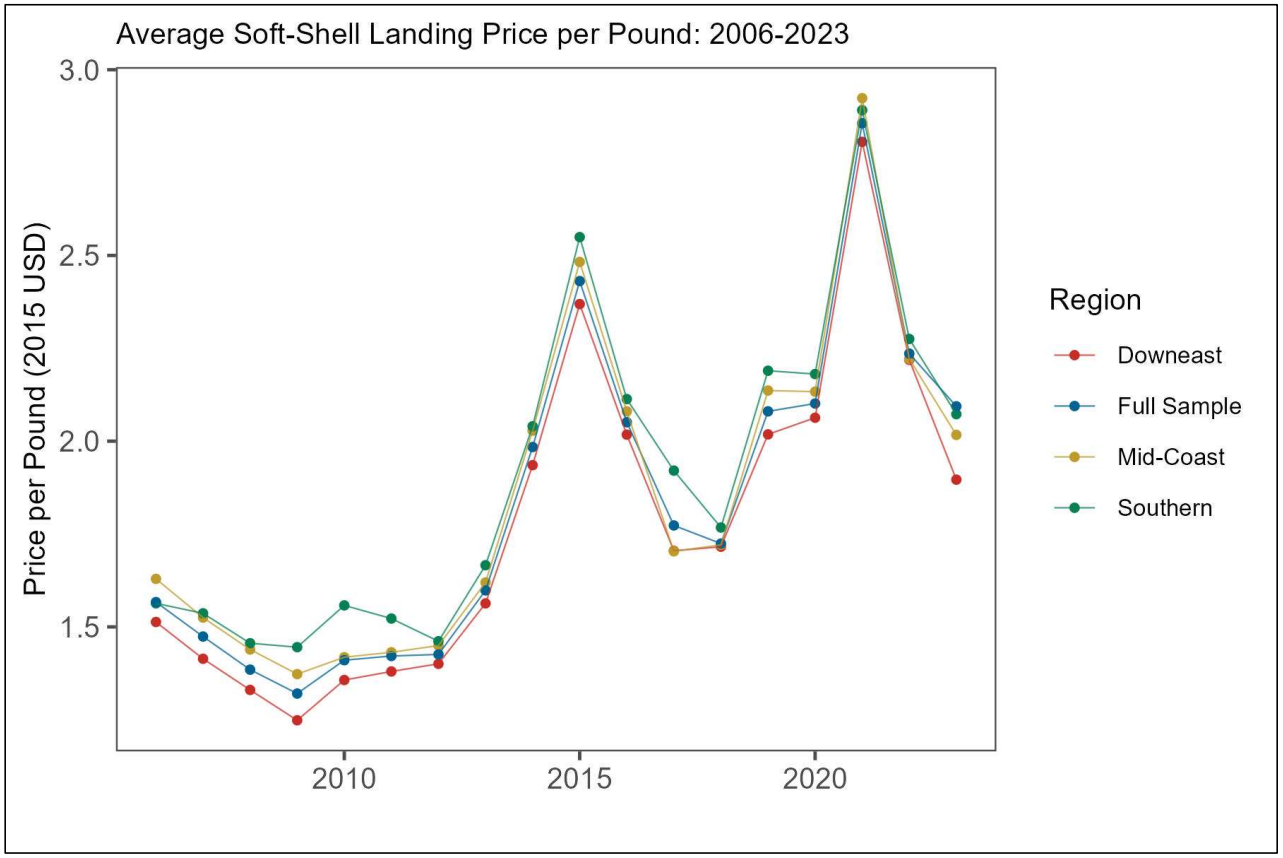
- Significant increase in average landing price per pound in 2021
  - This has appeared to level out to pre-pandemic levels
- Regional Patterns Remain Consistent Across Time
  - Southern > Mid-Coast > Downeast
  - Gap between these regions varies by year
- More Formal Statistical Analysis to Follow



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Next Steps + Additional Thoughts

- Compute weekly price indices at both a state and regional level
  → Examine sub-regional patterns?

- Examine landings and ex-vessel price patterns to investigate environmental, macroeconomic, and other socioeconomic factors that influence them
  → Would like feedback on which factors to prioritize

  - Green Crab Predation
  - Harmful Algal Blooms
  - Pollution/Water quality
  - Ocean warming
  - Water quality
  - Intertidal access
  - Co-management & stock enhancement activities
  - COVID-19